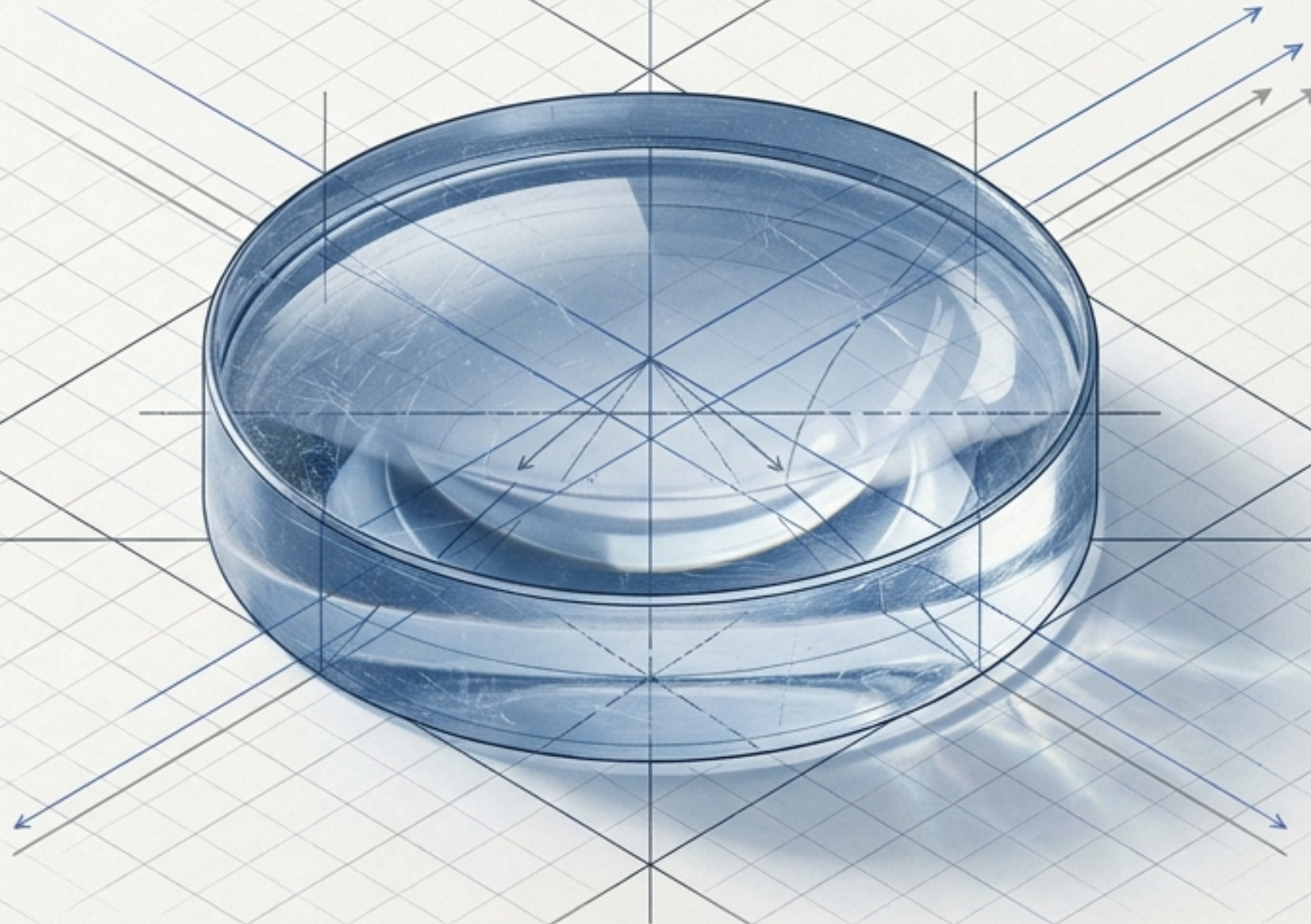
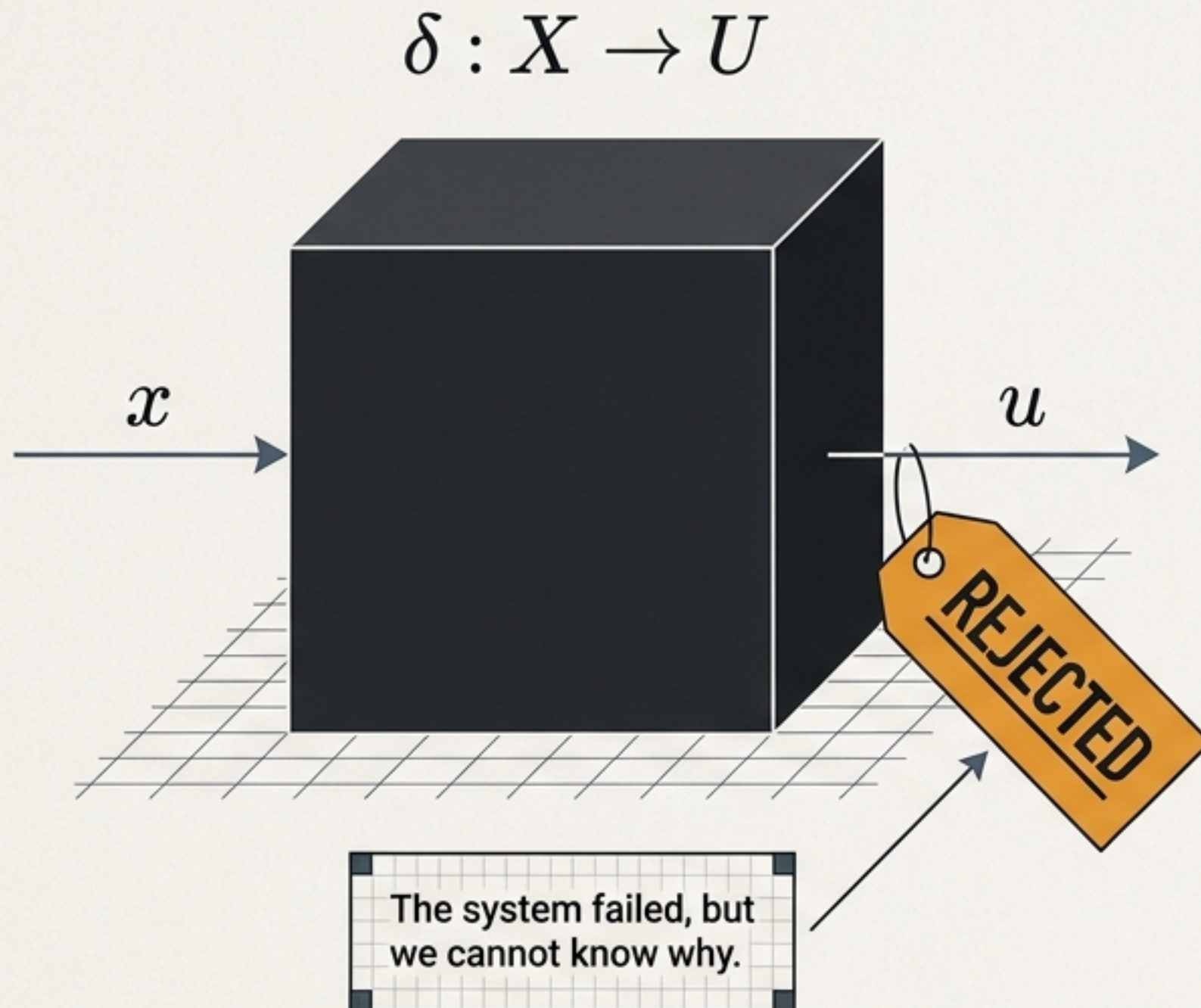




THE LAYERED DECISION ARCHITECTURE

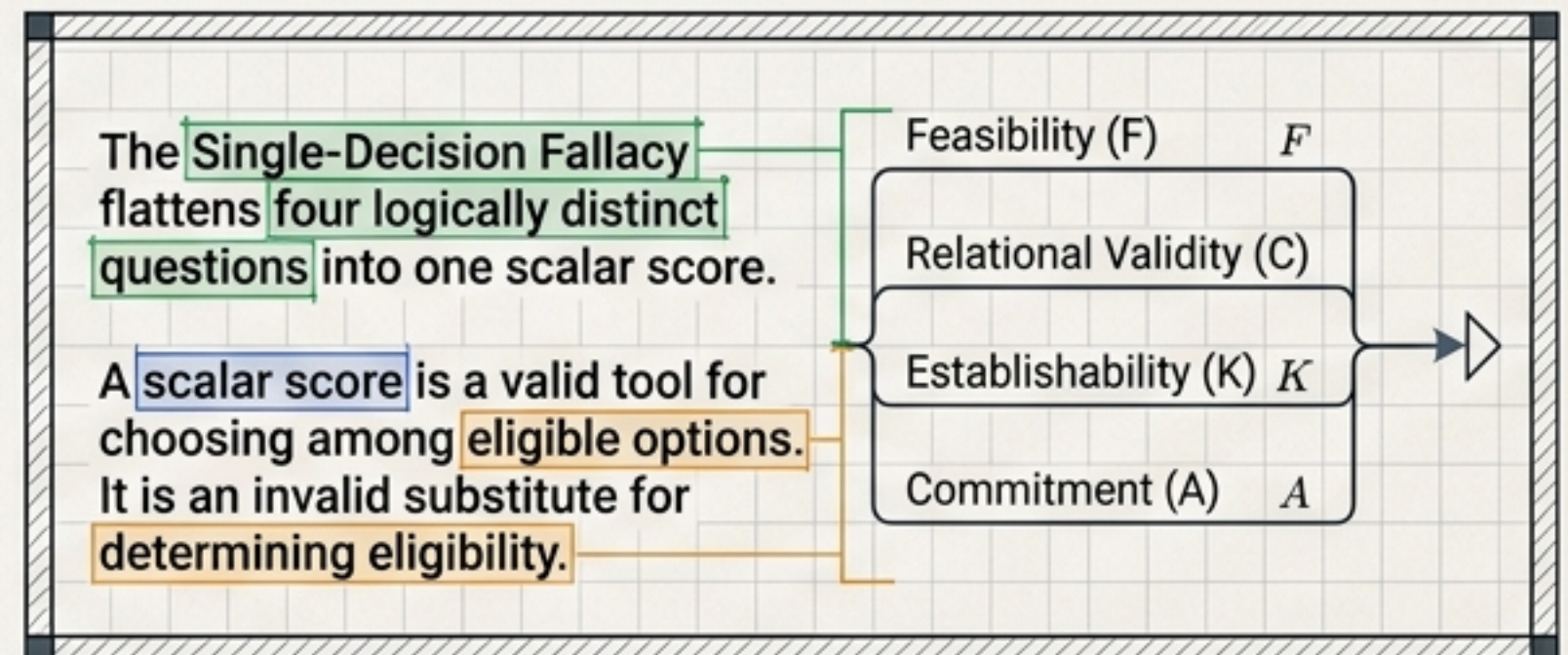
Feasibility, Relational Validity, Establishability, and Commitment in History-Dependent Decision Systems.
A structural playbook for engineering informative failures.

THE CONVENTIONAL MODEL



THE ALTERNATIVE

“Do not scalarize a distinction before determining whether the distinction is structurally relevant.”

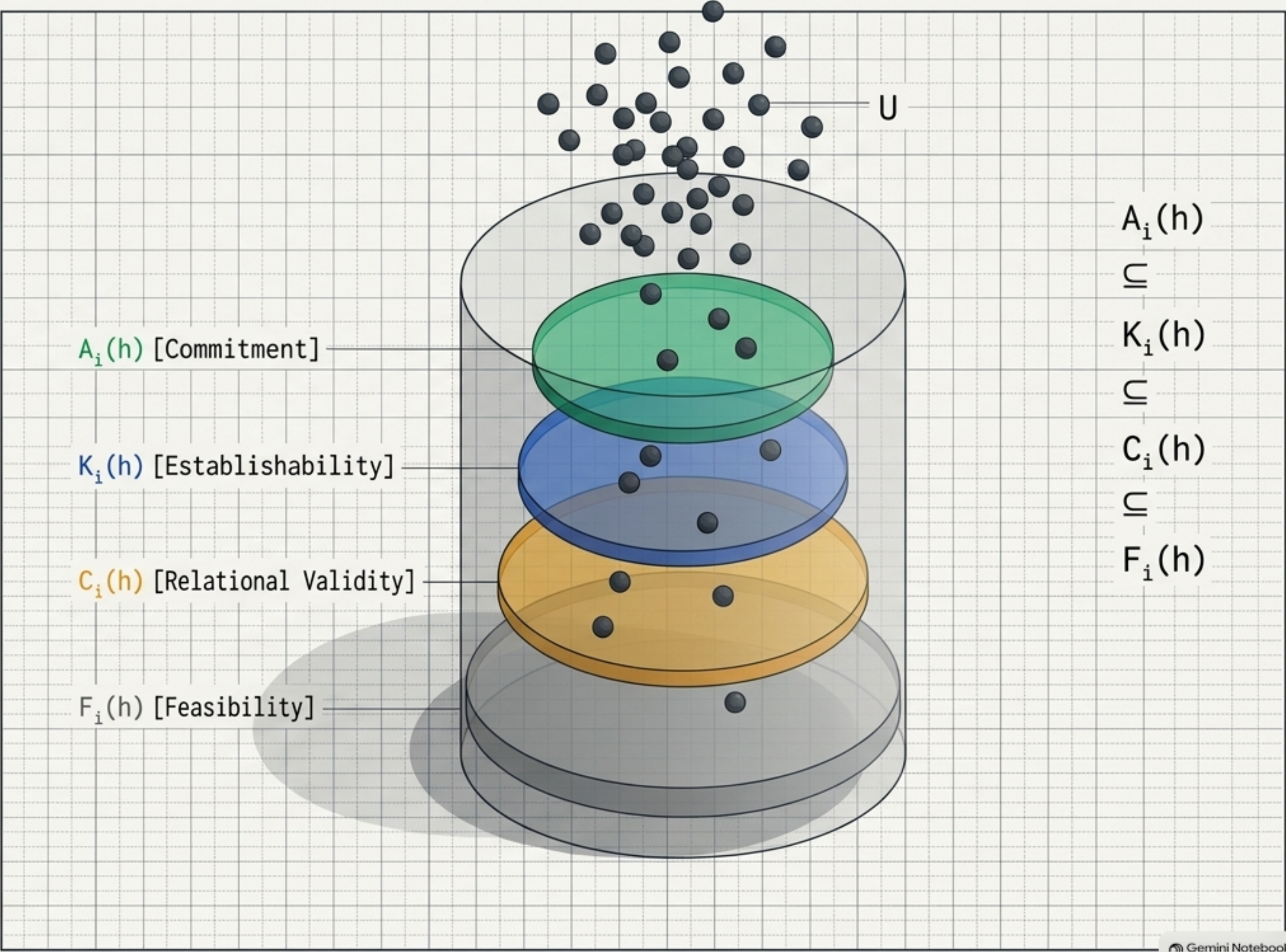


The anatomy of an informative decision

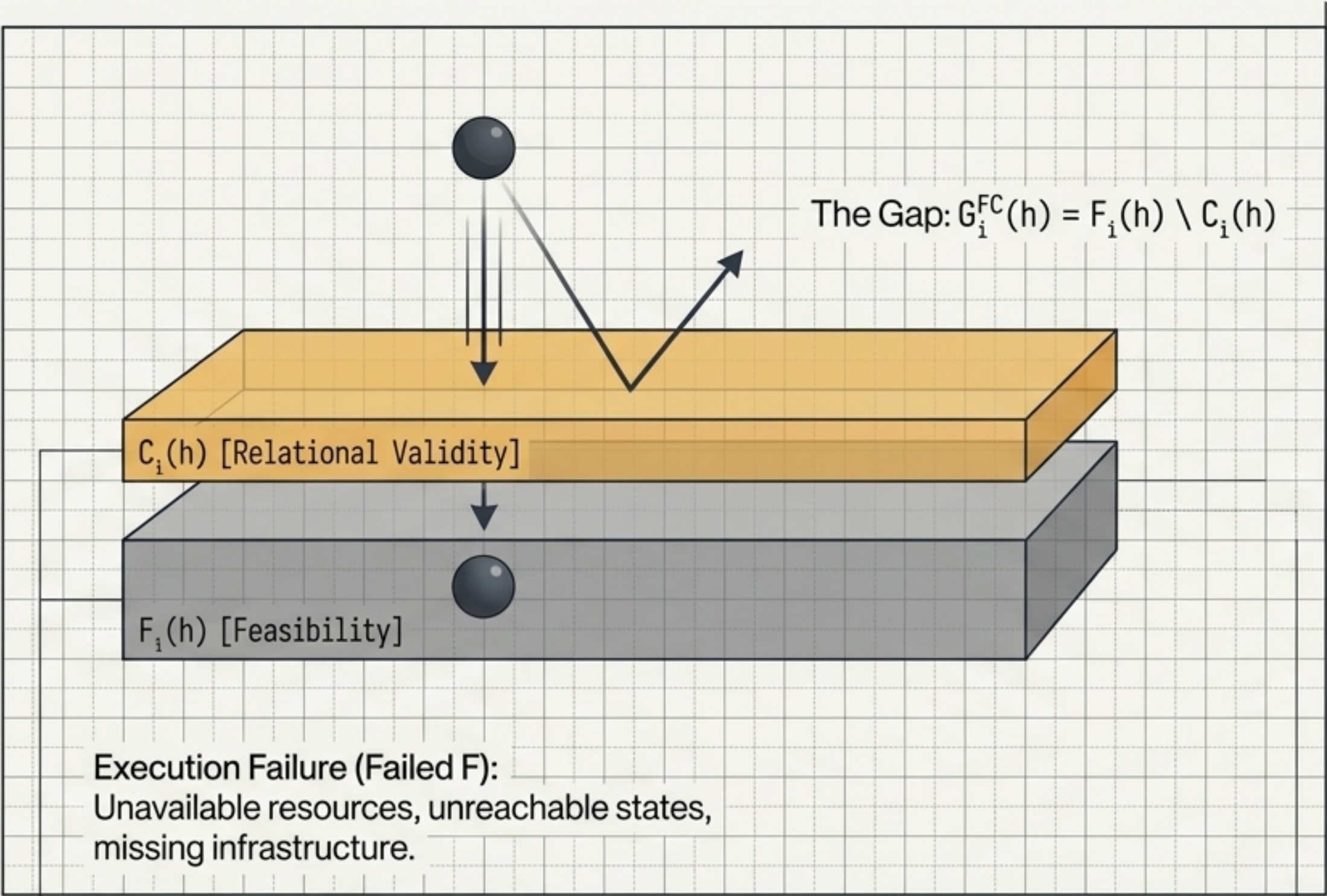
Instead of a single function, an action must survive a four-layer actor- and history-relative hierarchy.

- $F_i(h)$ [Feasibility]
- $C_i(h)$ [Relational Validity]
- $K_i(h)$ [Establishability]
- $A_i(h)$ [Commitment]

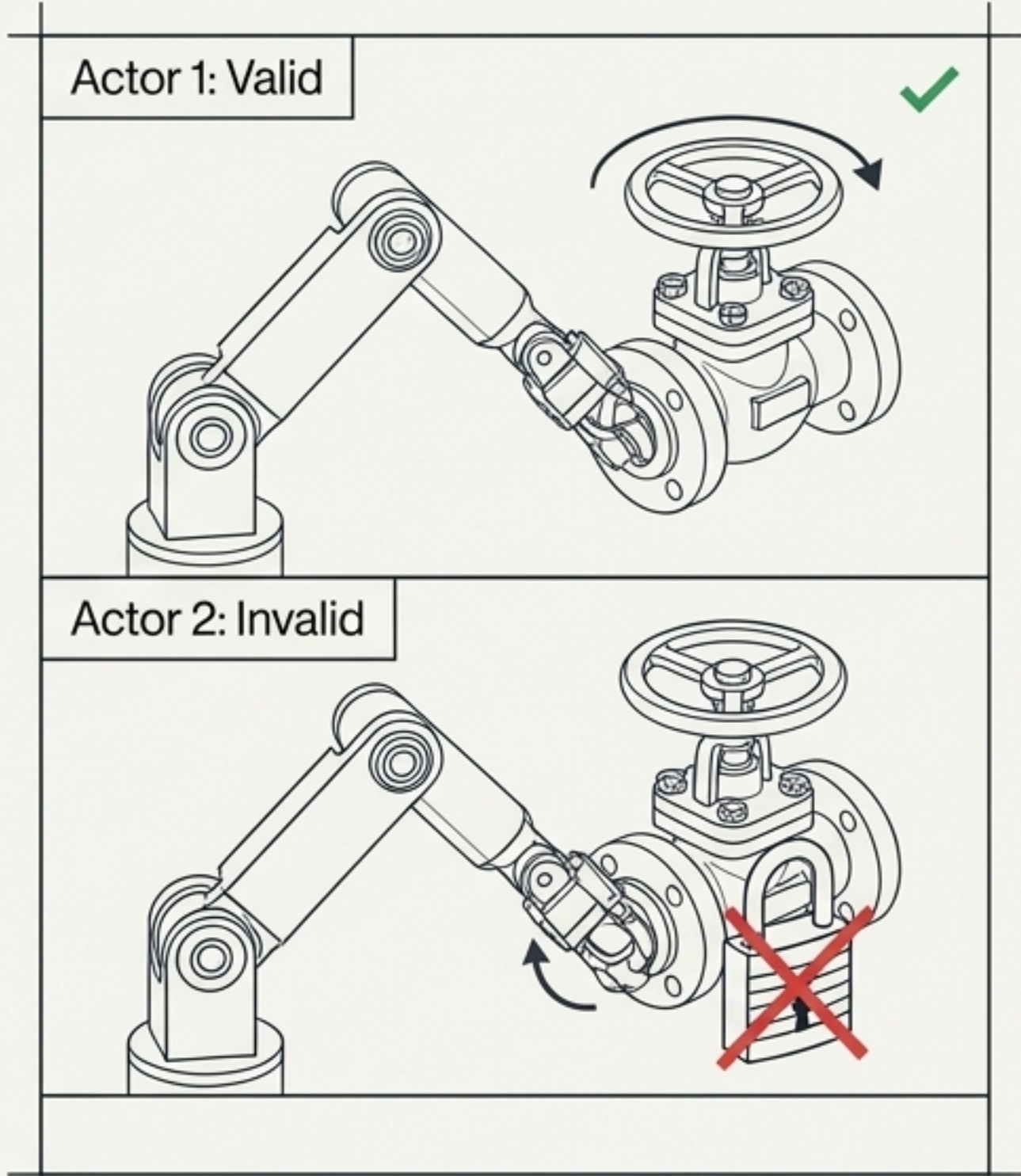
When an action is rejected, its failure location provides the exact architectural diagnosis.



Physical capability does not entail relational validity



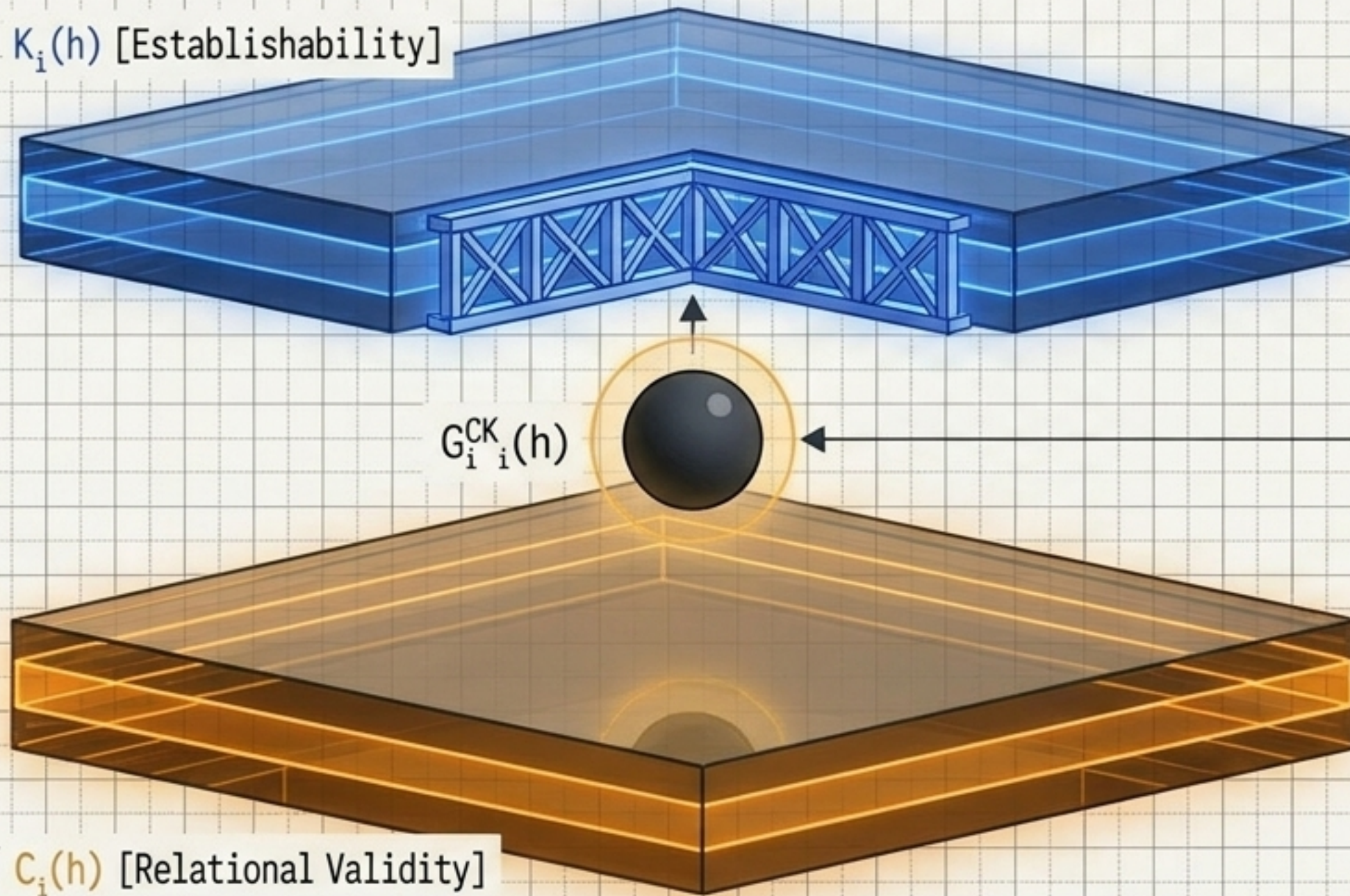
Relational Exclusion (Failed C):
The action is physically possible but violates
safety bounds, contracts, roles, or obligations.



Key Insight: State-transition semantics alone
 $T(h, u) = T(h, v)$ cannot characterize relational status.

The epistemic gap between truth and proof

The Gap: $G_i^{CK}(h) = C_i(h) \setminus K_i(h)$



Epistemic Exclusion:
The action satisfies all governing relations, but the actor cannot establish this from its current information position.

Soundness over Completeness:
A perfectly sound system $K_i(h) \subseteq C_i(h)$ will frequently be **incomplete**. It will reject valid actions because it lacks the evidence object $e_i(h, u)$ required to certify them.

Optimization belongs exclusively in the commitment layer

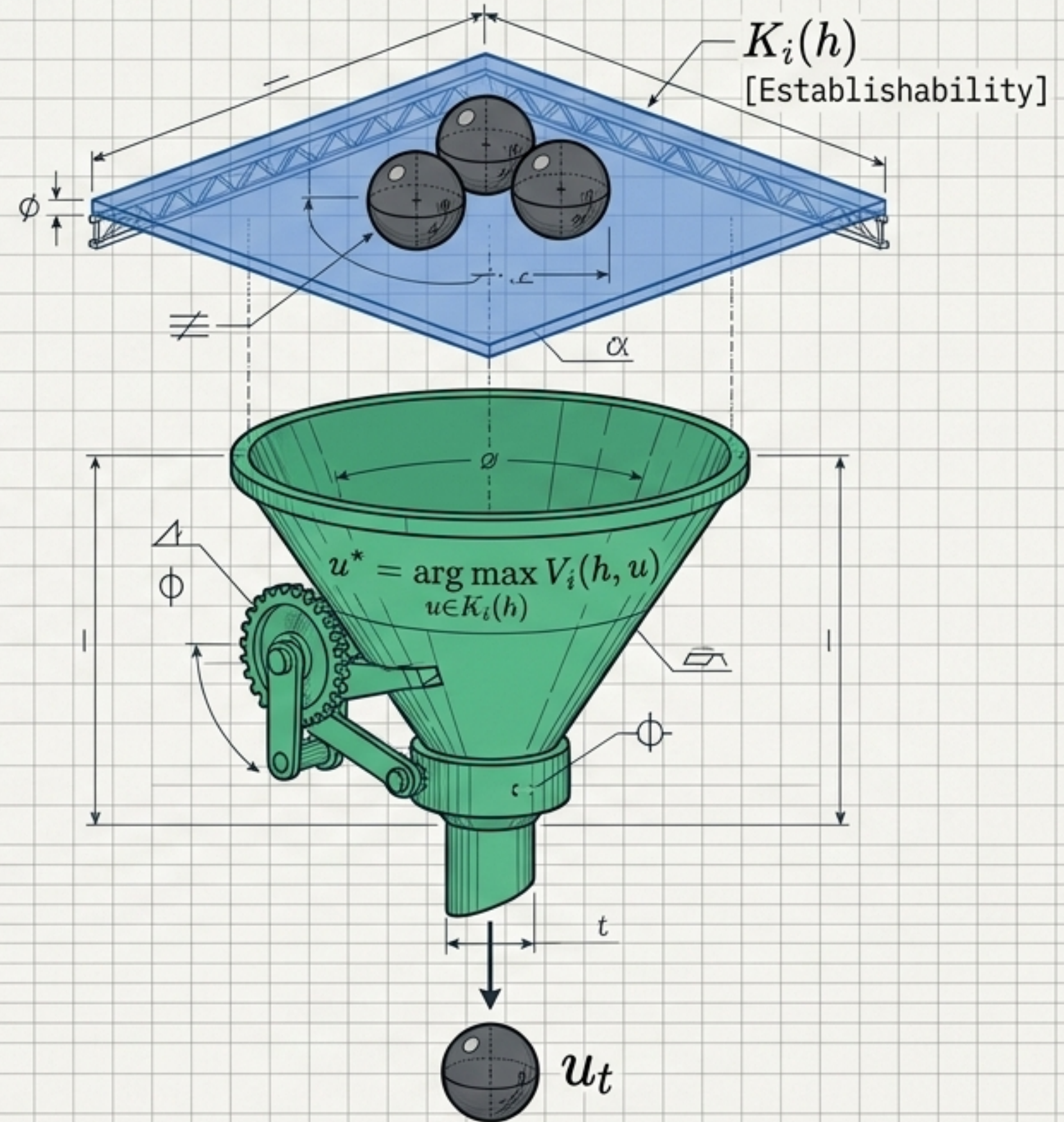
Optimization belongs exclusively in the commitment layer

The Gap: $G_i^{KA}(h) = K_i(h) \setminus A_i(h)$

Policy Rejection:

An action is executable, valid, and certifiable—but not selected by the policy π_i .

Scalar optimization is a selection mechanism. It must operate strictly inside the boundaries of what is already established, never as a substitute for the boundaries themselves.

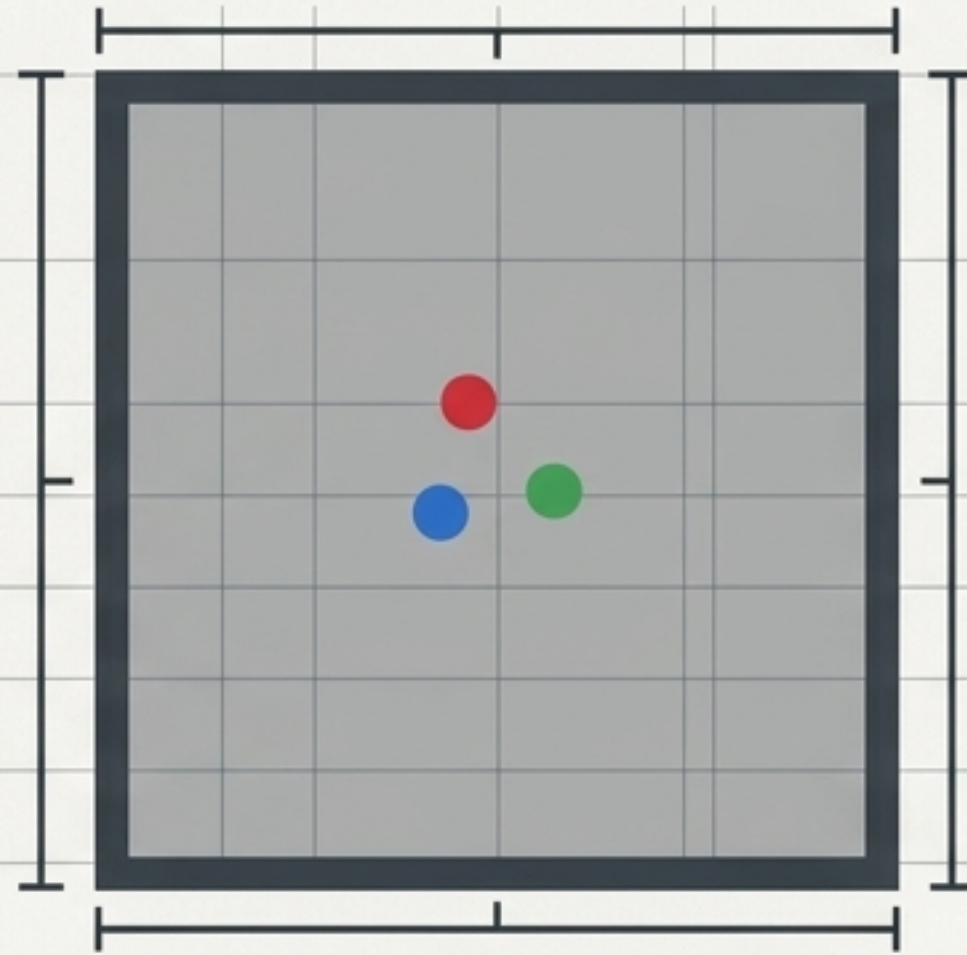


The non-implication calculus

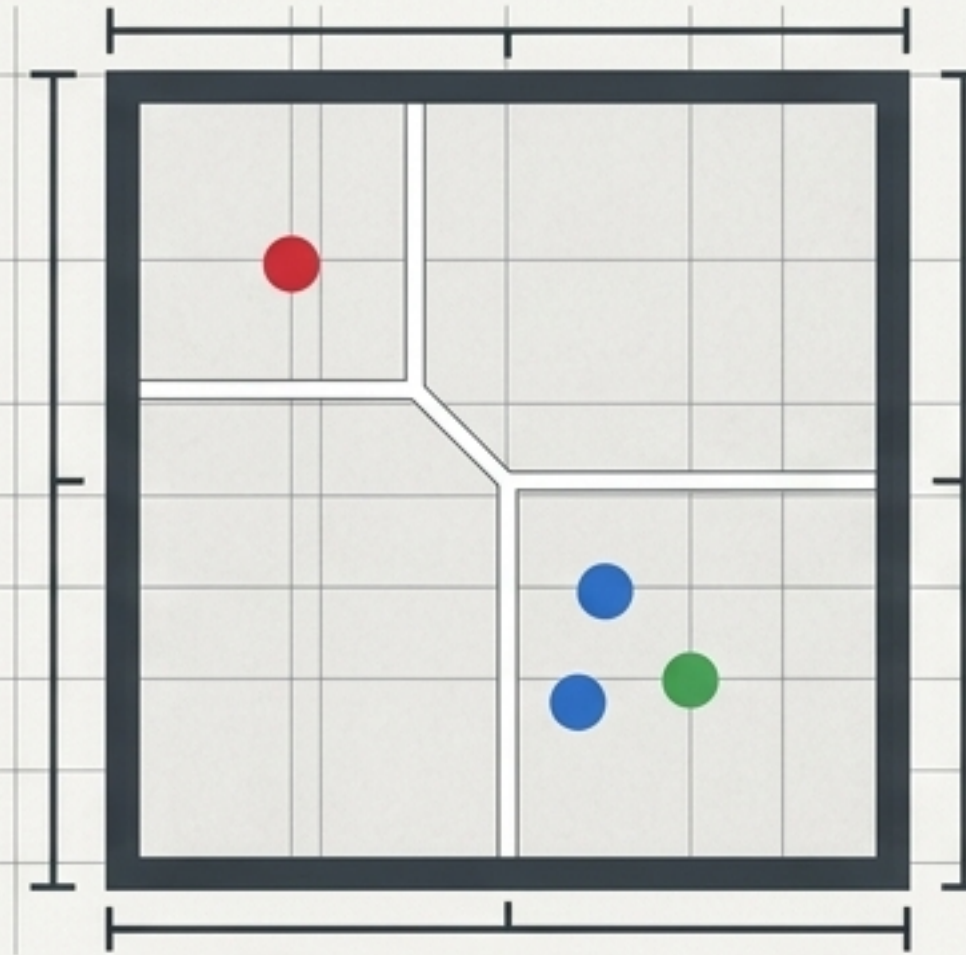
Tempting Inference		Structural Reality
$u \in F_i(h)$ I can execute it	$\Rightarrow$	$u \in C_i(h)$ It is safe/valid to do
$u \in C_i(h)$ It is valid to do	$\Rightarrow$	$u \in K_i(h)$ I have the evidence to prove it
$u \in K_i(h)$ I have proven it valid	$\Rightarrow$	$u \in A_i(h)$ It is the optimal choice
$h_t \subseteq h_{t+1}$ I have recorded more history	$\Rightarrow$	$K_i(h_t) \subseteq K_i(h_{t+1})$ I have more justified options
$D(x_1) = D(x_2)$ The states look identical	$\Rightarrow$	$Q(x_1) = Q(x_2)$ The states require the same action

Diagnostic design is equivalence-class engineering

Observation $O_i: X \rightarrow Y_i$ induces a partition of the state space.



Coarse Diagnostic



Refined Diagnostic

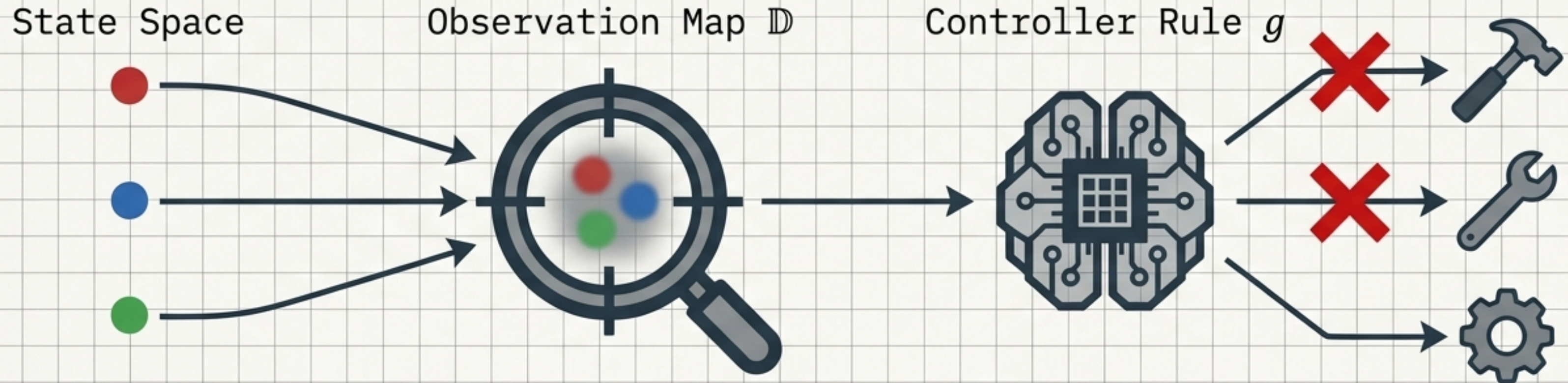
More sensors do not equal more information. Information only increases when a partition separates states requiring different downstream treatment.

The goal is a minimal sufficient observation: exactly tracking necessary distinctions—no coarser, and no finer.

The diagnostic impossibility theorem

If a representation $\mathbb{D}$ collapses two states that require different interventions $\mathbb{R}$, no downstream controller g can recover the distinction correctly.

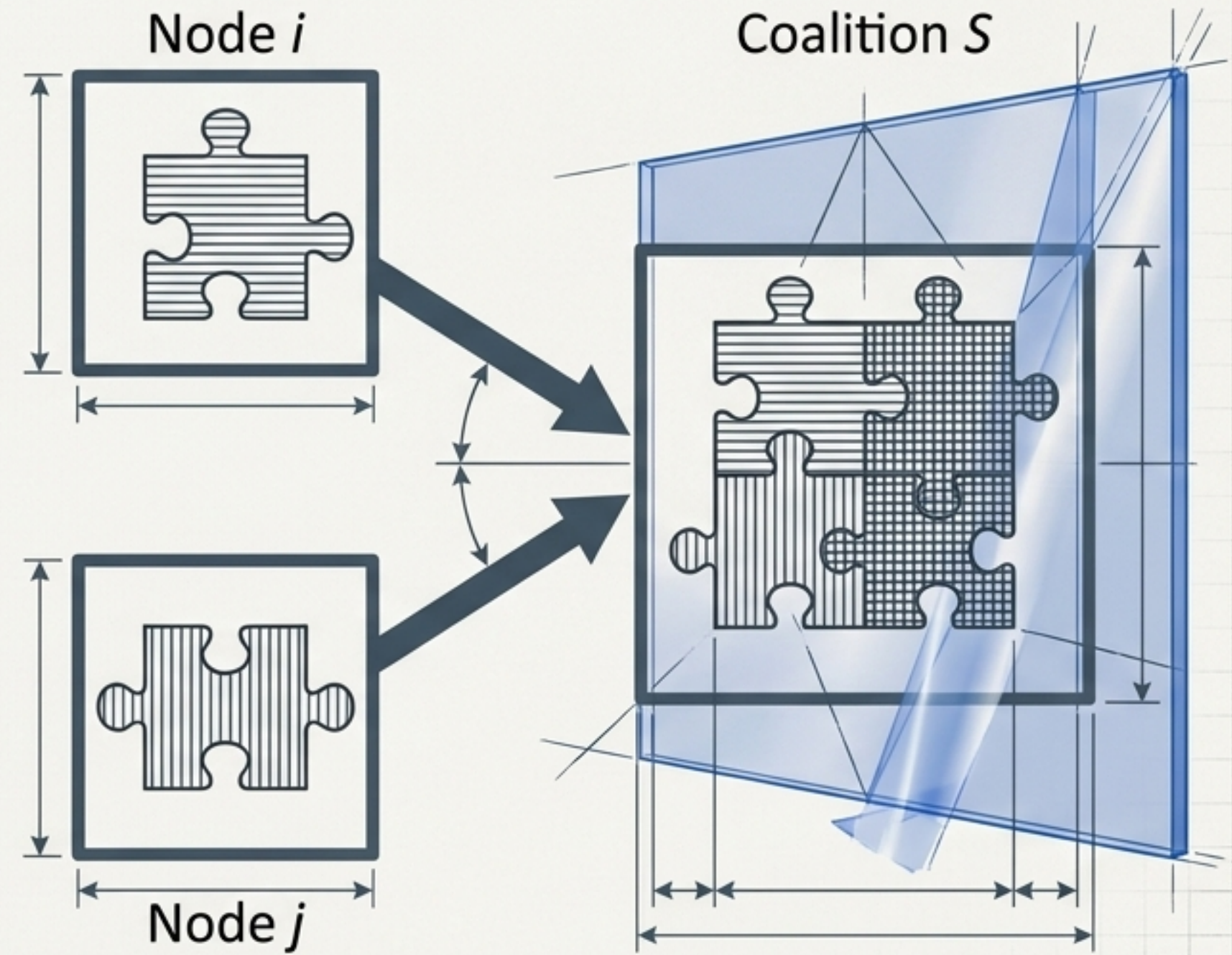
$$\mathbb{D}(\mathbf{x}_1) = \mathbb{D}(\mathbf{x}_2) \Rightarrow g(\mathbb{D}(\mathbf{x}_1)) = g(\mathbb{D}(\mathbf{x}_2))$$



The lesson: The defect is in the partition, not the policy.
You cannot compute your way out of a collapsed representation.

Knowledge composition is an evaluation position problem

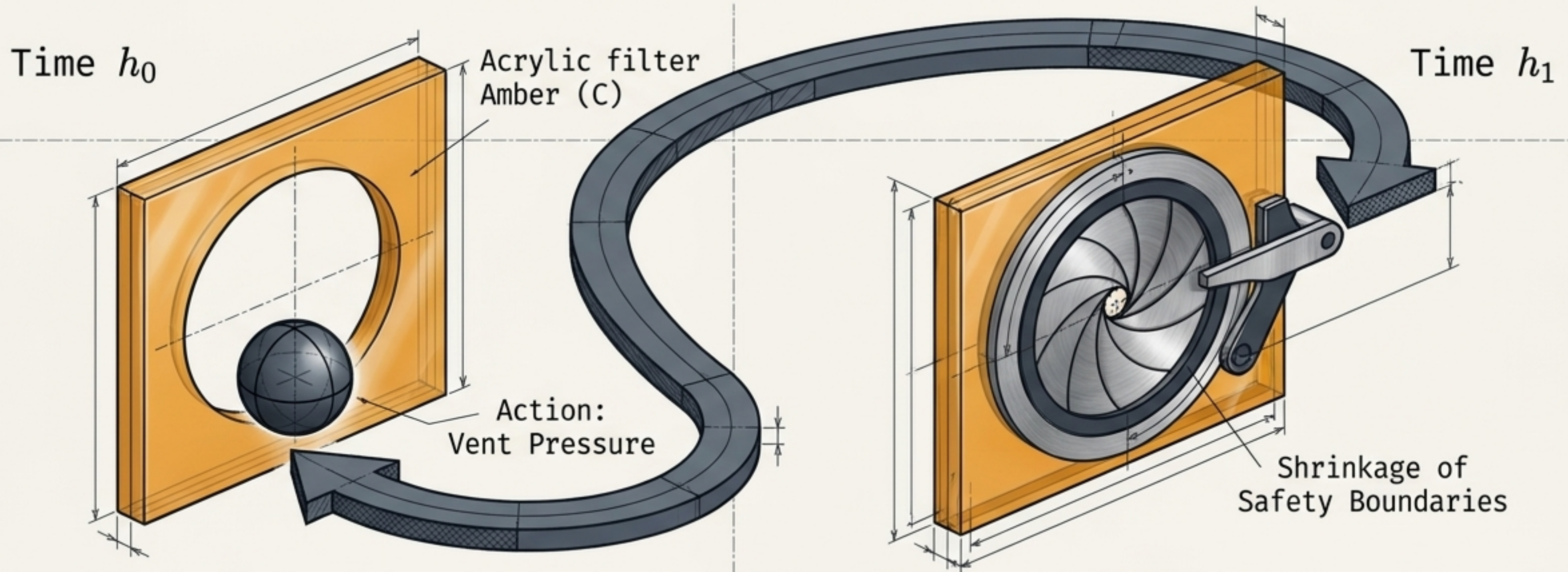
- Gap I: Fixed position failure. Requires refining the observer O_i in place.
- Gap II: Distributed information failure. Neither observer alone can distinguish the states.
- $K_i(h) \cup K_j(h) \neq K_{ij}(h)$
- The union of separate certifications is not joint certification.
- Closing the gap requires changing the evaluation position, composing evidence $e_i \oplus e_j$ into a genuinely new knowledge object.



History is non-monotonic: decisions reshape their own boundaries

$$h_t \subseteq h_{t+1} \not\Rightarrow K_i(h_t) \subseteq K_i(h_{t+1})$$

The transition from history h_t to h_{t+1} is mediated by the actor's own action.
Actions trigger new relational requirements, obligations, and safety locks. The accumulation of history can actively shrink the set of justified actions.

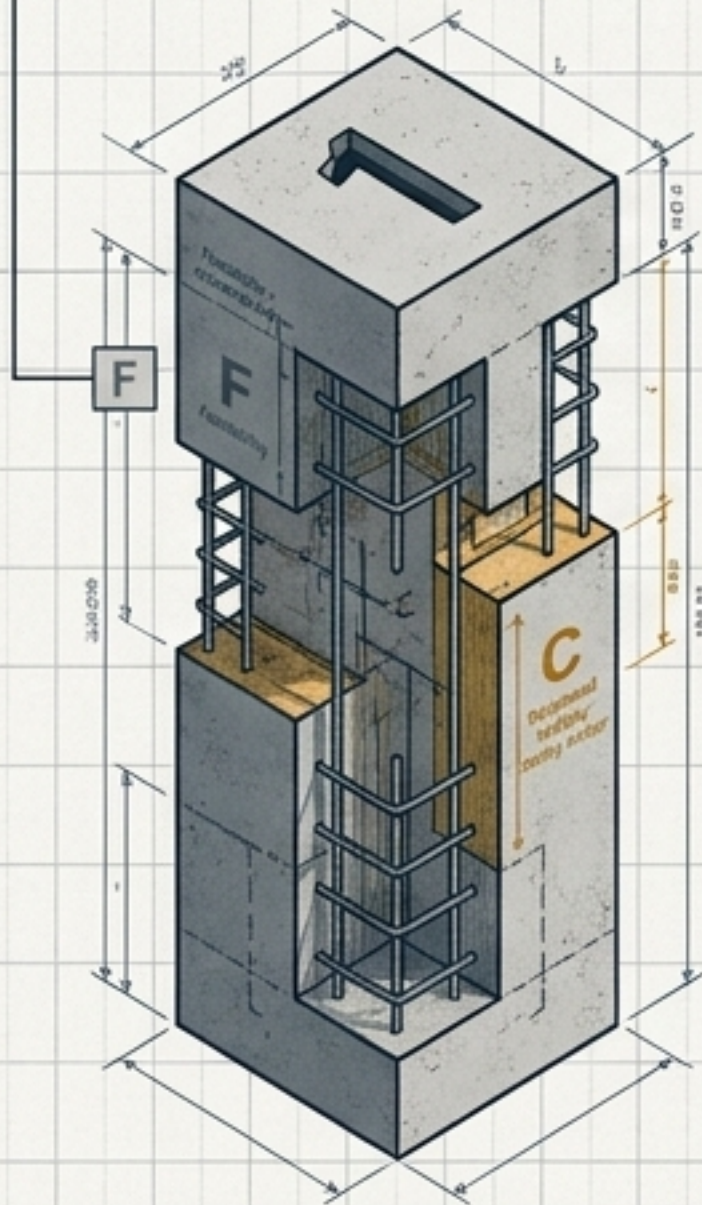


The diagnostic taxonomy of decision failures

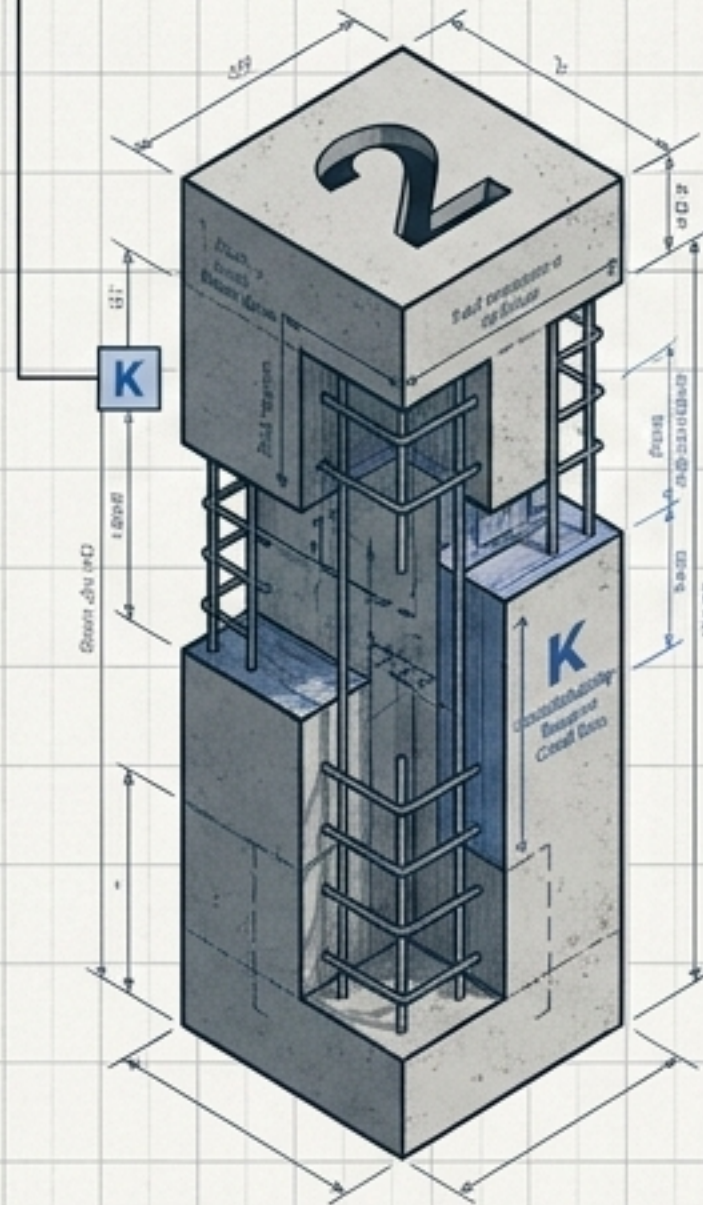
Type	Gap Equation	Description	Remedy
Type I (Execution)	$u \notin F_i$	Action physically blocked.	Resource allocation.
Type II (Relational)	$F_i \setminus C_i$	Executable but invalid.	Correct safety bounds/roles.
Type III (Epistemic)	$C_i \setminus K_i$	Valid but uncertifiable.	Upgrade sensors/evidence.
Type IV (Policy)	$K_i \setminus A_i$	Certifiable but rejected.	Revise optimization/goals.
Type V (Aliasing)	$D(x_1) = D(x_2)$	Ambiguous diagnostics.	Finer partition mapping.
Type VI (Position)	$K_i \subseteq K_S$	Missing distributed context.	Widen evaluation position.

Architectural design principles

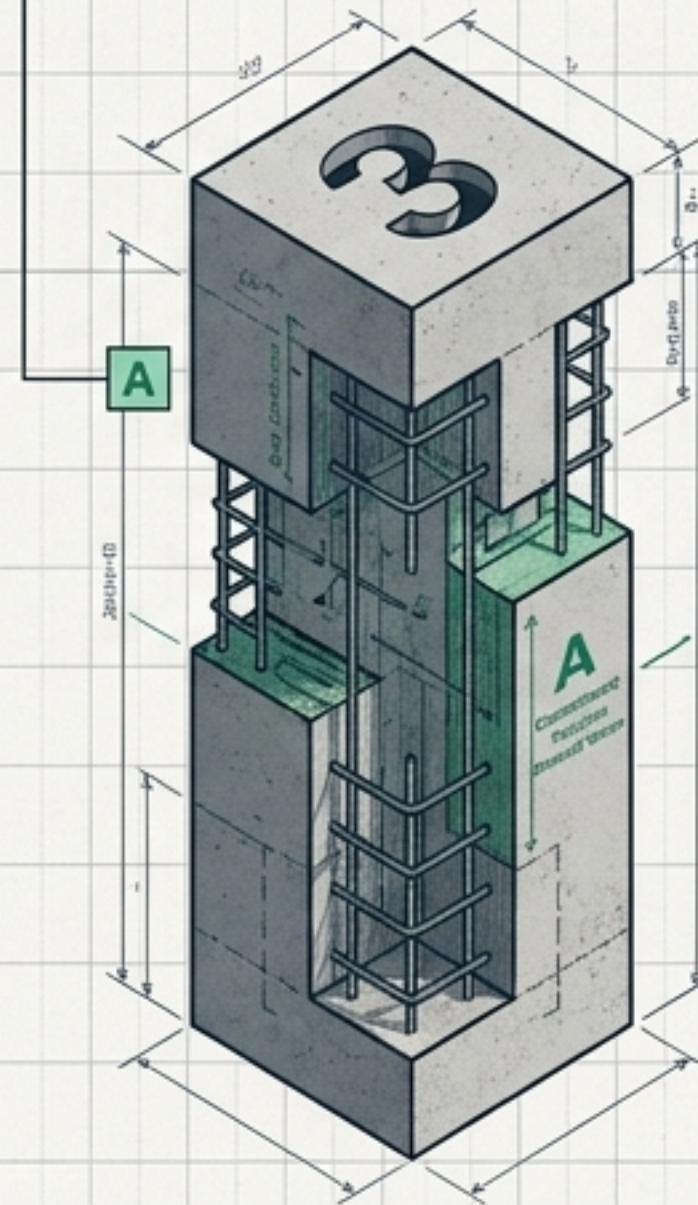
1. Preserve the Boundaries:
Never collapse F , C , K , and A .
A system that only reports
"action rejected" has erased the structural information required to fix it.



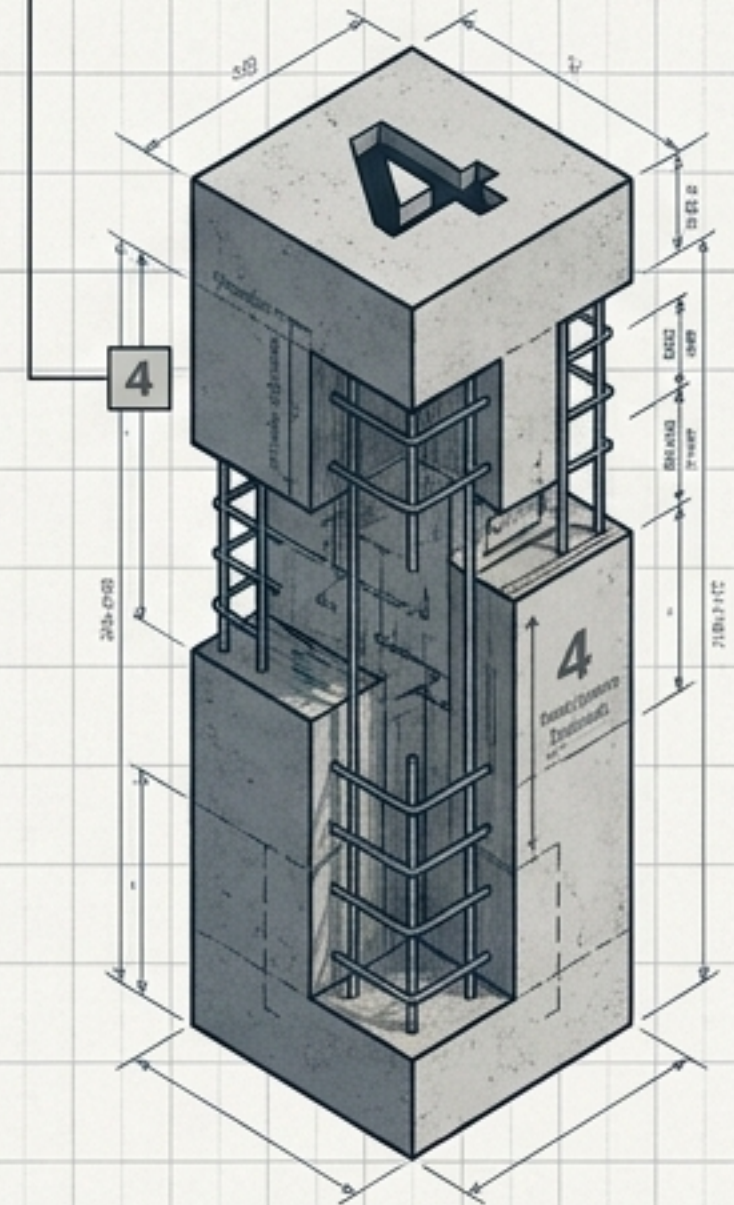
2. Treat Diagnostics as Partition Design: Stop maximizing sensor count.
Engineer the exact partition of the fault space that tracks the distinctions your interventions require.



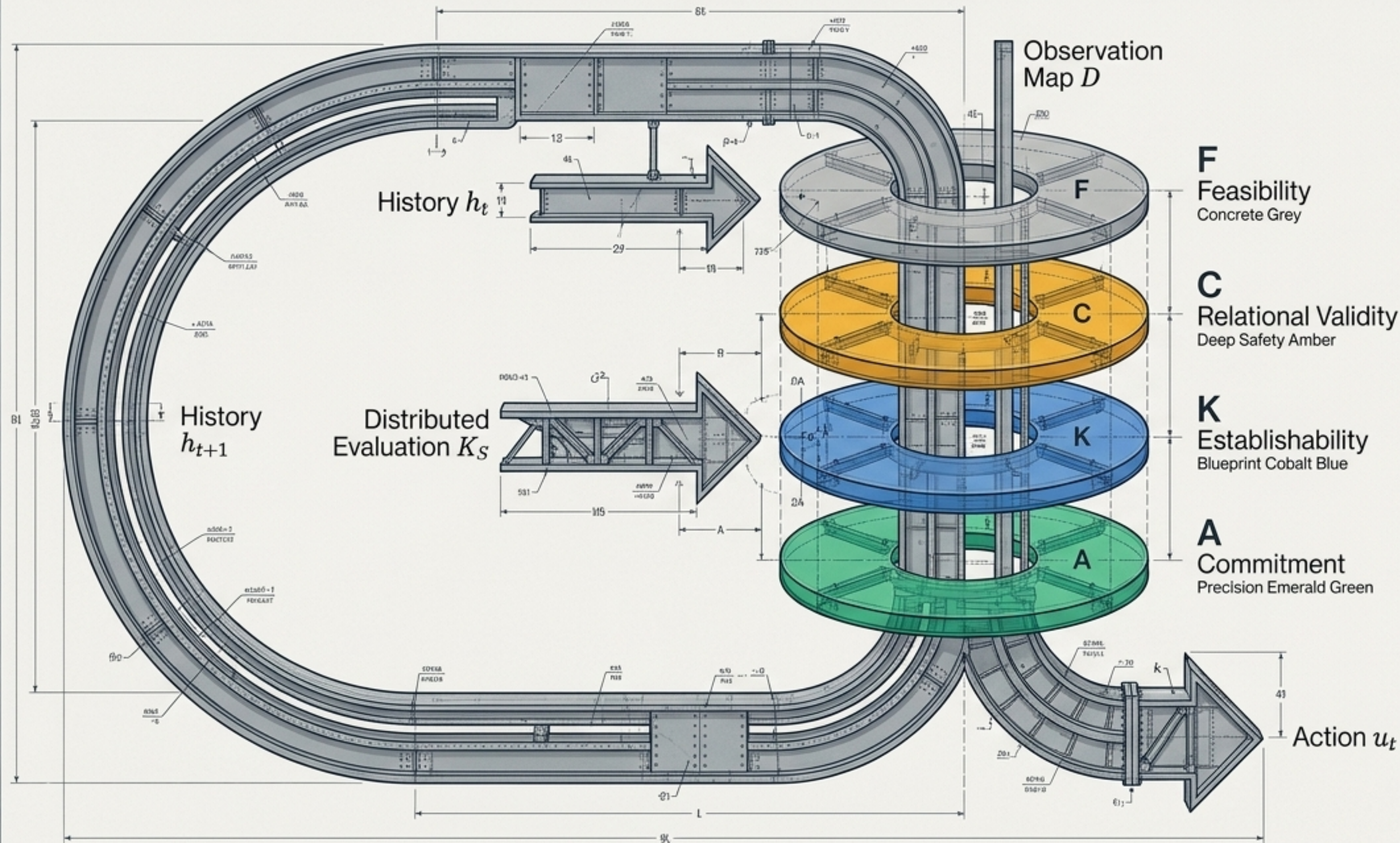
3. Change the Evaluation Position: When missing distinctions are outside an actor's information channel, compose positions rather than forcing local inference to guess.



4. Make History Explicit:
Treat history not as a container of accumulated facts, but as the active determinant of the relations governing future actions.



The layered decision engine



Can do $\neq$
Satisfies $\neq$
Can establish $\neq$
Should commit.

Every layer is indexed by history.

Every transition feeds back into the relations governing the next round.

This is the architecture of decision adequacy.